

Roll No. 220010256023

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**B. Pharma. 6th Semester
(Regular/Re-appear/Improvement)**

Examination – May, 2025

PHARMACEUTICAL BIOTECHNOLOGY

Paper : BP605T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory* consisting of ten questions carrying 2 marks each. Section-B contains nine questions carrying 5 marks each and students have to attempt any *seven* questions. Section-C contains *three* questions carrying 10 marks each and students have to attempt any *two* questions.

SECTION – A

1. Attempt *all* questions.

10 × 2 = 20

(a) What is Protein Engineering ?

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- (b) Enlist some applications of biosensors in pharmaceutical industries.
- (c) Enumerate uses of DNA ligase and restriction endonucleases.
- (d) Describe the significance of multiple cloning sites in cloning vector.
- (e) What are hypersensitivity reactions ?
- (f) Write down the applications of hybridoma technology.
- (g) Write the principle of Sandwich ELISA.
- (h) What is frameshift mutation ?
- (i) Enlist *four* equipments used in fermentation.
- (j) Enumerate uses of dried human plasma.

SECTION – B

Note : Attempt any *seven* questions $5 \times 7 = 35$

- 2. Discuss basic principles of genetic engineering.
- 3. Describe any *two* methods of enzyme immobilization.
- ✓ 4. Explain the production and industrial application of Lipase.
- 5. Write an exhaustive note on recombinant DNA technology.

6. Discuss the importance of PCR technique.
7. Describe method of preparation of toxoids.
8. Write a short note on structure and function of MHC.
9. Differentiate genome organization of Eukaryotes and Prokaryotes.
10. Write a note on collection, processing and storage of whole human blood.

SECTION – C

Note : Attempt any *two* question :

2 × 10 = 20

11. Discuss principle, various steps and applications of western blotting.
12. Discuss the production of penicillins by fermentation.
13. Write short notes on any *two* :
 - (i) Aeration process in fermentation
 - (ii) Plasma substitutes
 - (iii) Structure of immunoglobulins

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B.Pharma 6th sem 1st Sessional Examination, 2025
Paper Code- BP605T, Pharmaceutical Biotechnology

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

- a. Define Biotechnology to Pharmaceutical science $2 \times 5 = 10$
- b. Define Enzyme immobilization and one application
- c. Principle of genetic engineering and protein engineering
- d. Define Working of biosensors and application
- e. Define cloning vectors

2 Attempt any one questions from the rest $10 \times 1 = 10$ marks

- 1.a. Describe Brief introduction to PCR
- 2.b. application of r-Dna technology and production of genetic engineering 1.vaccines-hepatitis and interferon

3 Attempt any two questions from the rest $5 \times 2 = 10$ marks

- a. Explain Biosensors working and application in pharmaceutical industries
- b. Explain Method of enzyme immobilization and application
- c. Explain restriction endonucleases with example